

STUDENT ACTIVITY FILE — INNOVATIVE DESIGN LAB (IDL) - I

SENTINEL: Autonomous Hybrid-AI SOC Analyst & Investigative Engine

Academic Year: 2026-2027 | Lead Innovator: SIVABALAN T & Team | Institution: Sairam Institutions

SECTION 0: COVER PAGE & METADATA SHEET

Field	Details
Project Title	SENTINEL (Security Event Network Triage Investigation with Neural Engine & LLM)
Project ID	SAIRAM-IDL1-2026-CYBER-09 (Version 1 - New Project)
Team Members	1. SIVABALAN T (2026-CSE-CYBER-01) - Lead Innovator 2. Team Member 2 (2026-CSE-CYBER-02) - Cyber Specialist 3. Team Member 3 (2026-CSE-CYBER-03) - Full-Stack Developer
Primary SDG	SDG 9: Industry, Innovation, and Infrastructure (SAP-SDG9-CYBER-01)
Secondary SDG	SDG 16: Peace, Justice, and Strong Institutions (SAP-SDG16-SEC-02)
Tertiary SDG	SDG 8: Decent Work and Economic Growth (SAP-SDG8-AUTO-03)
Competitions	Hac'KP 2026 @ Zoho Corporation Campus (In-Person Pitching Round)

WEEK – 1: ORIENTATION & ENTREPRENEURIAL MINDSET

1. Innovation & Entrepreneurship Analysis:
 - **Core Problem:** Security analysts suffer from severe alert fatigue (10,000+ daily logs). Commercial AI tools (ChatGPT/Claude) cannot be used directly because corporate/government data leaks to cloud servers.
 - **Innovation:** SENTINEL invents Reversible Tokenized Pseudonymization — scrubbing 100% of sensitive PII locally before AI reasoning occurs.
 - **Value Proposition:** Delivering enterprise-grade autonomous triage at \$0 operational cost for 90% of routine alerts on local consumer GPUs (NVIDIA RTX 3050).
2. Growth Mindset & Market Size:
 - **TAM / SAM:** \$13.2 Billion+ Global SIEM Automation & Threat Intel Market (18.5% CAGR).
 - **Target Customers:** Enterprise SOC's, MSSP's, Air-Gapped Government Labs, Cyber Police Cells, Startups.
 - **Entrepreneurial Traits:** Quantizing 70B models down to 4-bit (IQ3_M) to run flagship AI on edge hardware.

■ WEEK – 2: PROBLEM IDENTIFICATION & PASSION CV

1. Real-World Societal Need:

Small businesses, hospitals, and public institutions cannot afford \$100k+/year enterprise SIEM subscriptions (like Splunk or Palo Alto Cortex XSOAR), leaving them vulnerable to ransomware. SENTINEL provides an open-source, self-hosted, autonomous AI SOC co-pilot running on standard workstation hardware.

2. Passion CV — SIVABALAN T (Lead Innovator):

- **Technical Specialization:** Cybersecurity & SIEM (Wazuh, Linux Hardening, MITRE ATT&CK;), AI Engine (Ollama, GGUF Quantization Q4_K_M/IQ3_M, 3-Tier Router, ChromaDB RAG), Zero-Trust Privacy Engineering.
- **Project Experience:** Lead Innovator of SENTINEL (reduced triage from 45 mins to < 30s); Hac'KP 2026 National Finalist @ Zoho; HackTronix 2.0 Vision & Agent Track.
- **5-Year Vision:** Publish IEEE research, launch SENTINEL as an open-source enterprise engine on GitHub, and establish a deep-tech cybersecurity venture.